

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI-590 018



A REPORT ON MINI-PROJECT WORK
SMART FM RADIO RECEIVER USING ARDUINO NANO
Submitted in the partial fulfillment of the requirements for the award of the degree of
BACHELOR OF ENGINEERING
in
ELECTRONICS & COMMUNICATION ENGINEERING

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**DEPARTMENT OF ELECTRONICS & COMMUNICATION
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Certificate

Certified that the Mini-Project work entitled “**Smart FM Radio Receiver using Arduino Nano**” carried out by, Mr. Tilak G P, USN 4BD22EC108, Ms. Vaishnavi B N, USN 4BD22EC110, Mr. Vedamurthy A N, USN 4BD22EC112, Mr. Venkatesh R Shettar, USN 4BD22EC114, Bonafide students of this institution submitted in partial fulfillment for the award of degree of **Bachelor of Engineering in Electronics & Communication** by **Visvesvaraya Technological University, Belagavi** during the academic year **2024-25**. It is certified that all corrections / suggestions indicated for continuous internal evaluation have been incorporated in the report deposited in the department library. The report has been approved as it satisfies the academic requirements in respect of Mini-Project Work prescribed for the said degree.

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This course will enable us to:

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- Acquire technical knowledge and collect the information.
- Enhance communication, technical presentation and report preparation skills.
- Provide an opportunity to exercise the creative and innovative ideas in group.

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ABSTRACT

Radio technology has evolved from traditional analog systems like AM and FM radios, which rely on manual tuning, to advanced digital platforms such as Digital Audio Broadcasting (DAB), HD Radio (High-Definition Radio), and satellite radio, offering higher sound quality and additional features. Internet radio further enhances accessibility by streaming content online, while software-defined radios (SDRs) provide flexibility through software-driven functionality. Hybrid radios combine traditional broadcast with internet connectivity for a seamless listening experience. Despite these advancements, many systems lack integration between manual controls and modern digital interfaces, highlighting the need for more user-friendly and versatile designs.

With the growing demand for customizable and user-friendly devices, there is a need for an FM radio system that combines traditional functionality with modern digital interfaces. Such a system would cater to users seeking simplicity while providing advanced feature like remote control via mobile device.

This project aims to design a Smart FM radio receiver capable of tuning into a frequency range of 87.0 MHz to 107.0 MHz, with user-friendly control over frequency and volume through potentiometers. It also includes an interactive LCD display interface to provide information of the selected frequency, radio station, and volume settings.

This project is programmed through the Arduino IDE for seamless implementation. For remote control functionality, the Electronics Bluetooth mobile application is used, enabling users to interact with the system via virtual buttons for volume and station adjustments.

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CHAPTER 1

INTRODUCTION

1.1 History of Radio

The history of radio dates back to the late 19th century and is marked by a series of key developments that transformed communication and entertainment.

1. **Early Foundations** (1860s-1890s):

The theoretical groundwork for radio was laid by James Clerk Maxwell, who predicted the existence of electromagnetic waves in the 1860s. In the 1880s, Heinrich Hertz demonstrated the practical existence of these waves by successfully transmitting and receiving electromagnetic signals, proving Maxwell's theories.

2. **The Birth of Wireless Communication** (1890s):

Guglielmo Marconi, an Italian inventor, is widely credited with the invention of radio as a practical means of communication. In 1895, Marconi transmitted the first wireless signal across a distance of 1.5 miles, and by 1901, he had successfully transmitted a wireless signal across the Atlantic Ocean. This marked the dawn of radio communication.

3. **The First Radio Broadcasts** (1920s):

The 1920s saw the first commercial radio broadcasts, transforming radio from a point-to-point communication tool into a mass medium for entertainment and information. In 1920, the first radio station, KDKA, began broadcasting in Pittsburgh, Pennsylvania, primarily for news and music. This decade also witnessed the rise of Amplitude Modulation (AM) broadcasting, which became the standard for radio transmission.

4. **The Rise of FM Radio** (1930s-1940s):

In the 1930s, Edwin Armstrong developed Frequency Modulation (FM) radio, which offered better sound quality and less interference than AM radio. FM radio became more popular in the 1940s and 1950s, especially for music broadcasting, due to its superior audio fidelity.

5. **The Golden Age of Radio** (1930s-1950s):

Radio became the primary form of mass entertainment during the 1930s to the 1950s. It was a vital medium for news, entertainment, and advertising. Radio dramas, news programs, and music shows dominated the airwaves, and the invention of portable radios made it more accessible to the public.

6. **The Digital Era** (1990s-present):

In the 1990s, digital radio technologies such as Digital Audio Broadcasting (DAB) and HD Radio emerged, offering clearer sound and additional features like station metadata. The rise of

satellite radio and internet radio further expended the reach and diversity of radio content, providing global access to music, news, and talk shows.

Today, radio continues to evolve, integrating with digital platforms and streaming services, while maintaining its importance as a tool for communication, entertainment, and information.

1.2 Radio Transmitter

A radio transmitter is a device that generates and sends electromagnetic signals (radio waves) over a specific frequency range to be received by radio receivers. It is a key component in wireless communication systems, enabling the transmission of audio, video, or data. The basic function of a radio transmitter is to encode information onto a carrier wave and radiate it through an antenna.

Components of a Radio Transmitter

1. **Oscillator:** The oscillator generates the carrier wave, a continuous electromagnetic signal at a specific frequency. This is the foundational signal on which the information (audio, data, etc.) will be modulated. The frequency of the carrier is crucial as it determines the communication channel for the transmitter.
2. **Modulator:** The modulator is responsible for encoding the information onto the carrier wave. It varies one of the properties of the carrier wave, such as its amplitude, frequency, or phase, to represent the information being transmitted.
 - AM varies the amplitude of the carrier wave to encode the audio signal.
 - FM changes the frequency of the carrier wave to encode the signal, offering better sound quality and resistance to interference.
 - PM changes the phase of the carrier signal to encode the information.
3. **Amplifier:** The amplifier boosts the modulated carrier signal to a power level suitable for transmission over long distances. The strength of the transmitted signal determines its range, with higher power amplifiers enabling longer transmission distances.
4. **Antenna:** The antenna radiates the amplified signal as electromagnetic waves. The design and size of the antenna are determined by the frequency of the signal and the desired range. The antenna's purpose is to efficiently radiate the signal into the air, making it detectable by radio receivers.

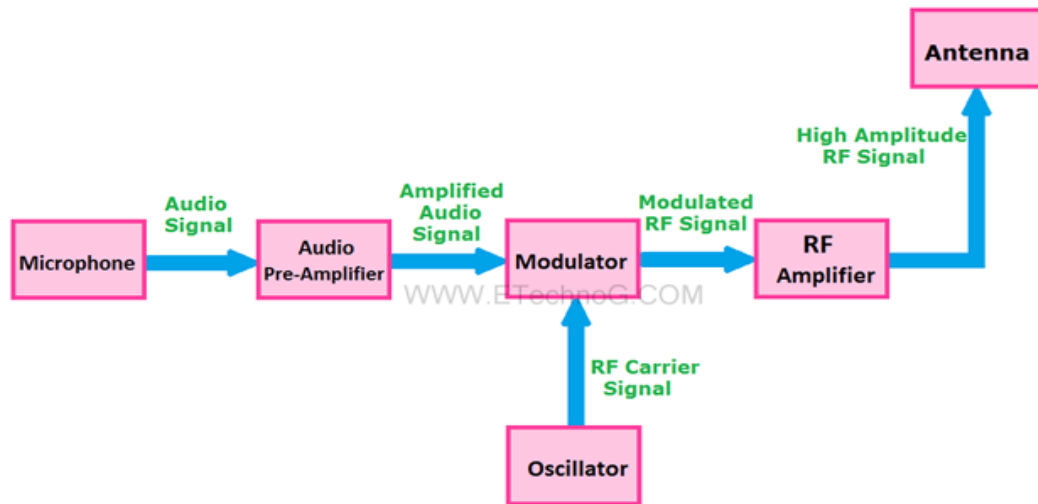


Figure 1.2.1 Block diagram of FM Transmitter

Types of Radio Transmitters

1. AM Radio Transmitter: Use amplitude modulation to vary the strength (amplitude) of the carrier wave.
2. FM Radio Transmitter: Use frequency modulation to vary the frequency of the carrier wave.
3. Digital Radio Transmitter: Use digital modulation techniques such as Quadrature Amplitude Modulation (QAM) and Phase Shift Keying (PSK) to encode information more efficiently, allowing for better sound quality and more channels.
4. Satellite Radio Transmitter: Transmit signals to satellites, which then relay the signal to receivers on the ground.

Applications of Radio Transmitter

- Broadcasting: Transmitting AM, FM, or digital radio signals to the public for news and entertainment.
- Communication System: Used in two-way radios, mobile phones, and wireless communication systems.
- Satellite Communication: Used to send signals from Earth to satellites for global communication.
- Navigation: Radio transmitters are used in systems like Global Positioning System (GPS) to send signals for location tracking.

Radio transmitters are essential for any form of wireless communication, from traditional radio broadcasting to modern digital communication systems.

1.3 Radio Receiver

A radio receiver is a device that captures electromagnetic signals transmitted by a radio transmitter and converts them into usable information, such as audio, data, or images. It plays a critical role in communication systems, enabling users to access broadcasts or two-way communication. The receiver must extract the desired signal from a range of signals present in the airwaves and process it for output.

Components of a Radio Receiver

1. **Antenna:** The antenna captures electromagnetic waves from the air. Its design and size depend on the frequency of the incoming signal. It converts these waves into electrical signals for further processing.
2. **RF Tuner:** The tuner selects the desired frequency while rejecting all others. It consists of filters and oscillators to focus on a specific frequency band, ensuring only the intended signal is processed.
3. **Amplifier:** The amplifier boosts the strength of the weak signal received by the antenna, making it suitable for further processing without significant loss or distortion.
4. **Demodulator:** The demodulator extracts the original information (audio, data, etc.) encoded on the carrier wave. The type of demodulator used depends on the modulation technique (AM, FM, or digital).
5. **Audio/Signal Processor:** After demodulation, the extracted signal is processed for clarity and quality. In audio applications, this stage removes noise and prepares the signal for playback.
6. **Speaker or Output Device:** The processed signal is converted into a usable form, such as sound through a speaker or data displayed on a screen.

Types of Radio Receivers

1. **AM Receiver:** Designed to process amplitude-modulated signals. These are commonly used in broadcasting for news and talk shows due to their ability to cover long distances.
2. **FM Receiver:** Built to process frequency-modulated signals, providing superior sound quality and better resistance to noise. They are popular for music and entertainment broadcasts.
3. **Digital Radio Receiver:** Receive and process digital signals, such as those used in DAB (Digital Audio Broadcasting) and HD Radio. These receivers offer features like enhanced sound quality, additional channels, and metadata display.
4. **Satellite Radio Receiver:** Capture signals transmitted from satellites, allowing access to diverse channels globally. These are commonly used in automotive and home entertainment systems.
5. **Software-Defined Radio (SDR) Receiver:** Use software to process radio signals, providing flexibility to handle multiple modulation techniques and frequencies.

Working Principle of a Radio Receiver

1. **Signal Reception:** The antenna captures electromagnetic signals, including the desired and undesired frequencies.
2. **Frequency Selection:** The tuner isolates the desired frequency, rejecting other signals and noise.
3. **Signal Amplification:** The selected signal is amplified to a level suitable for demodulation.
4. **Demodulation:** The encoded information is extracted from the carrier wave.
5. **Signal Processing:** The extracted information is enhanced for clarity and quality.
6. **Output:** The processed signal is delivered as sound (audio), data, or other forms.

Types of Modulation Techniques Used in Radio Signal Transmission

Modulation is the process of encoding information onto a carrier wave to transmit it efficiently over a distance. The carrier wave's properties—amplitude, frequency, or phase—are varied based on the input signal.

The key modulation techniques used in radio signal transmission:

1. Amplitude Modulation (AM)

- **Principle:** Varies the amplitude of the carrier wave while keeping its frequency and phase constant.
- **Application:** Used in AM radio broadcasting, aviation communication, and long-range communication.
- **Advantages**
 - Simple to implement.
 - Effective for long-distance transmission.
- **Disadvantages**
 - Prone to noise and interference.
 - Lower sound quality compared to other techniques.

2. Frequency Modulation (FM)

- **Principle:** Varies the frequency of the carrier wave according to the amplitude of the input signal, while amplitude and phase remain constant.
- **Application:** Commonly used in FM radio broadcasting, audio systems, and television sound transmission.
- **Advantages**
 - Better sound quality and resistance to noise compared to AM.
 - Higher fidelity in music and voice transmission.
- **Disadvantages**
 - Requires more bandwidth than AM.
 - Equipment is more complex.

3. Phase Modulation (PM)

- Principle: Varies the phase of the carrier wave in proportion to the amplitude of the input signal.
- Application: Used in digital communication systems and as a base for other modulation techniques like Quadrature Phase Shift Keying (QPSK).
- **Advantages**
 - High noise immunity.
 - Suitable for data transmission.
- **Disadvantages**
 - Complex implementation.
 - Phase changes may cause signal distortion.

4. Digital Modulation Techniques

Digital modulation is used in modern communication systems to encode digital data onto carrier waves. Common techniques include:

- Amplitude-Shift Keying: Encodes data by varying the amplitude of the carrier wave.
 - Used in low-bandwidth systems like Radio Frequency Identification (RFID).
- Frequency-Shift Keying: Encodes data by varying the frequency of the carrier wave.
 - Common in telemetry and radio data communication.
- Phase-Shift Keying: Encodes data by changing the phase of the carrier wave. Variants include:
 - Binary PSK (BPSK): One bit per phase shift.
 - Quadrature PSK (QPSK): Two bits per phase shift.
 - Used in satellite communication and Wi-Fi.
- Quadrature Amplitude Modulation: Combines amplitude and phase modulation for efficient data encoding.
 - Widely used in cable television and broadband internet.

5. Pulse Modulation Techniques

These are used for digital signals and involve sampling the input signal at discrete intervals. Common types include:

- Pulse Amplitude Modulation (PAM): Varies the amplitude of pulses.
- Pulse Width Modulation (PWM): Varies the width of pulses.
- Pulse Position Modulation (PPM): Varies the position of pulses.

Comparison table of types of modulation

Table 1.3.1 Comparison table of types of modulation

Modulation type	Noise Resistance	Bandwidth Usage	Complexity	Applications
AM	Low	Narrow	Simple	AM Radio, aviation
FM	High	Wider	Moderate	FM Radio, TV audio
PM	High	Wider	Complex	Data Communication
Digital	High	Efficient	High	Wi-fi, 4G/5G

1.4 FM Radio Receiver

An FM radio receiver is a system designed to receive and decode frequency-modulated (FM) signals, providing high-quality audio with better noise resistance compared to amplitude-modulated receivers.

Principle of FM Reception

FM receivers operate on the principle of capturing frequency-modulated signals and converting them into audio signals that can be played through a speaker. The carrier wave's frequency varies according to the amplitude of the input signal (e.g., voice or music), and the receiver detects these variations to reproduce the original audio.

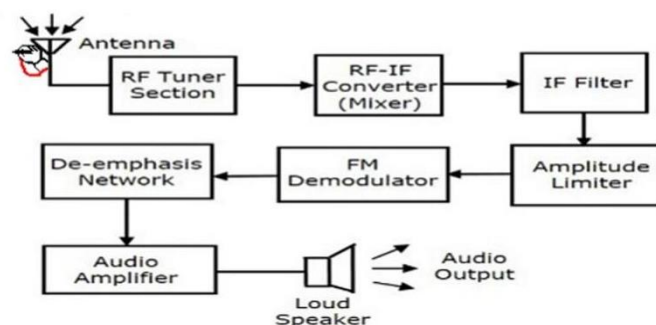


Figure 1.4.1 Block diagram of FM receiver

Block Diagram of FM Receiver

1. **Antenna:** Captures the electromagnetic waves from the air and converts them into electrical signals for processing.
2. **RF Amplifier:** Amplifies the weak RF signals received from the antenna and improves the signal-to-noise ratio.
3. **Tuner:** Selects the desired frequency (station) and rejects other frequencies and consists of bandpass filters and a local oscillator.
4. **Mixer and Local Oscillator:** Converts the received signal into an intermediate frequency signal for easier processing. It is achieved through a process called heterodyning.
5. **IF Amplifier:** Amplifies the intermediate frequency signal while maintaining its integrity and provides a stable and strong signal for demodulation.
6. **Demodulator (FM Detector):** Extracts the original audio signal by decoding the frequency variations in the modulated carrier wave.
7. **Audio Amplifier:** Boosts the audio signal to a level suitable for driving the speaker.
8. **Speaker:** Converts the amplified audio signal into sound waves for the listener.

Features of FM Radio Receivers

1. **Wide Bandwidth:** Offers better sound quality and fidelity compared to AM receivers.
2. **Noise Resistance:** FM signals are less affected by electrical noise and interference.
3. **Stereo Audio:** Modern FM receivers support stereo sound for a richer listening experience.
4. **Automatic Gain Control (AGC):** Ensures consistent audio output levels despite varying signal strengths.
5. **Digital Tuning:** In advanced FM receivers, digital signal processing (DSP) enables precise frequency selection and additional features like RDS (Radio Data System).

Working Steps of an FM Receiver

1. **Signal Reception:** The antenna captures FM signals over the range of 87.0 MHz to 108.0 MHz.
2. **Frequency Selection:** The tuner filters and selects the desired frequency from a range of incoming signals.
3. **Frequency Conversion:** Mixer and local oscillator convert the signal to a fixed intermediate frequency (10.7 MHz).

4. **Signal Amplification:** The IF amplifier boosts the signal strength without distortion.
5. **Demodulation:** The FM detector extracts the original audio signal encoded in the frequency variations.
6. **Audio Output:** The audio amplifier enhances the signal, which is then converted into sound waves by the speaker.

CHAPTER 2

LITERATURE SURVEY

[1] Title: Digital FM Transceiver Design and Construction using Microcontroller and I2C Interfacing Techniques.

Authors: Md. Kaimujjaman, Md. Mehedi Islam, Rubayat Mahmud, Mimmun Sultana, Md. Mahabub Hossain

Published in: *International Journal of Recent Technology and Engineering (IJRTE)*.

Published on: January 2020

Abstract: Communication is very much important for every community and radio communication is one of the easiest and simple media to provide information. The project represents the full duplex communication mode of FM transceiver (Radio + Transmitter). This paper describes on how to build a short-range FM transmitter using microcontroller. The paper also describes on how to build a FM radio receiver using micro-controller. Through the FM transmitter we can transmit our voice and any audio signal while through the radio receiver listener can listen any FM broadcasting channel. This project is based on KTMicro IC KT0803K monolithic digital stereo FM transmitter and Phillips TEA5767 radio receiver module. The quality of transmission and radio reception can be improved by Digital Signal Processing and Intelligent Tuning System. The heart of this project is Arduino UNO micro-controller which controls slave devices by Inter-Integrated Circuit (I2C) communication. The transmitter and the receiver both supports digital tuning and Phase Locked Loop (PLL) concept is used for recovery of the signals. The transmitter of this digital FM transceiver can be used for making announcements in schools, colleges, industries, hospitals, and other places using a condenser mic amplifier circuit and share music with the FM listeners using audio input port of the FM transmitter module. The radio receiver of this transceiver is used for listening any local FM broadcasting channel. It can be also used in specialized application such as local area network. For the construction of the transreceiver it requires some small and low-cost external equipment's.

Remarks: The project involves creating a digital FM transceiver (combining an FM transmitter and receiver) using Arduino microcontrollers. The system aims to support short-range (50 meters) communication in small areas like classrooms or offices.

[2] Title: Smart Radio using Arduino**Authors:** Bandari Theja, K. L. Deepika, K. Pavani, K. Sushmitha**Published in:** *International Journal of Research Culture Society***Published on:** April 2020

Abstract: smart radio idea is been very innovative in contemporary times making it by using Arduino Nano a cost efficient durable and economical which is portable with less power consumption makes it more efficient application which is very popular in hobby projects and readily available in the local market. For reliable data transmission and reception, we use antenna and potentiometer for adjusting frequencies and also volume. Audio amplifier plays a vital role in amplifying the audio signals and giving better performance at the output end. Here we consider Arduino UNO module in combination with Nokia 5110 LCD module for viewing the desired frequencies and this all are achieved some encoding scheme for reducing effect of noise. Present approaches make use of display library and library for using these modules with Arduino board. This gives an economical and reliable application which is compact, portable smart radio with no noise.

Remarks: The project aims to build a cost-effective, low-power FM receiver and radio using accessible components like the Arduino, Nokia 5110 LCD, and standard radio modules.

CHAPTER 3

OBJECTIVES

1. To design an FM radio receiver capable of tuning into a frequency range of 87.0 MHz to 107.0 MHz,
2. To enable precise control of frequency and volume.
3. To interface an display to know information such as the selected frequency, radio station, and volume settings.

CHAPTER 4

METHODOLOGY

1. Selection of Components
 - Choose the essential components, including a microcontroller, radio module, LCD display, potentiometers, audio amplifier, Bluetooth module, and speaker.
2. Hardware Assembly
 - Assemble the hardware on a breadboard to allow flexibility in testing and adjustments of various components.
3. Microcontroller Programming
 - Follow a structured approach to programming the microcontroller:
 1. Initialization of libraries for external modules and peripherals.
 2. Implementation of frequency control functionality.
 3. Development of display functionalities for user interaction.
 4. Management of audio output signals.
4. Bluetooth Connectivity
 - Establish a connection between the Bluetooth module and a compatible mobile application for remote control.
5. Testing
 - Conduct comprehensive testing to ensure that all components and functionalities work as intended.
6. Optimization and Final Adjustment
 - Perform noise reduction, amplification adjustments, fine-tuning, and software enhancements to improve overall performance and reliability.

4.1 BLOCK DIAGRAM

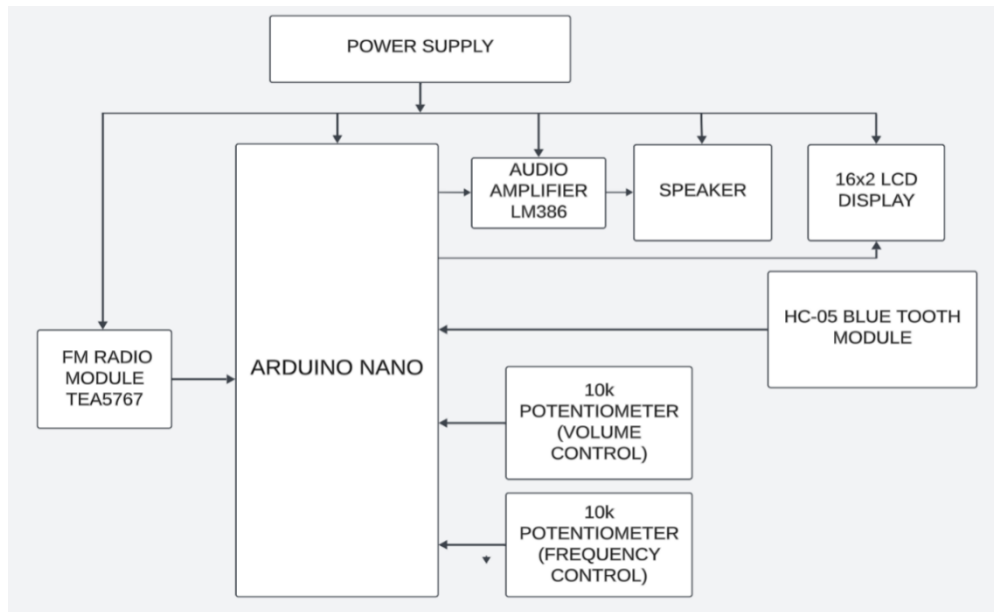


Figure 4.1.1 Block Diagram of Smart FM Radio Receiver using Arduino Nano

4.1.1 Block Diagram Description

- **Power Supply:**
Provides power to all components in the system, including the Arduino Nano, FM Radio Module, LCD Display, Bluetooth Module, and Audio Amplifier.
- **Arduino Nano:**
Acts as the main control unit for the system.
Interfaces with all the components, including the FM Radio Module (TEA5767), Audio Amplifier (LM386), LCD Display, Potentiometers, and HC-05 Bluetooth Module.
- **FM Radio Module (TEA5767):**
Receives FM radio signals.
Communicates with the Arduino Nano to tune to specific frequencies and process station data.
- **10k Potentiometers:**
Two potentiometers are used for:
 - **Volume Control:** Adjusts the audio output level.
 - **Frequency Control:** Tunes to different radio frequencies.
- **Audio Amplifier (LM386):**
Amplifies the audio signal from the FM Radio Module.
Sends the amplified signal to the speaker.
- **Speaker:**
Outputs the audio of the FM Radio.

- **16x2 LCD Display:**

Displays information such as the current frequency or station name.

Provides feedback to the user.

- **HC-05 Bluetooth Module:**

Enables wireless communication with a mobile device.

Allows users to control the smart radio operations (e.g., volume, frequency, or station change) via virtual buttons on their mobile application.

4.2 FLOW CHART

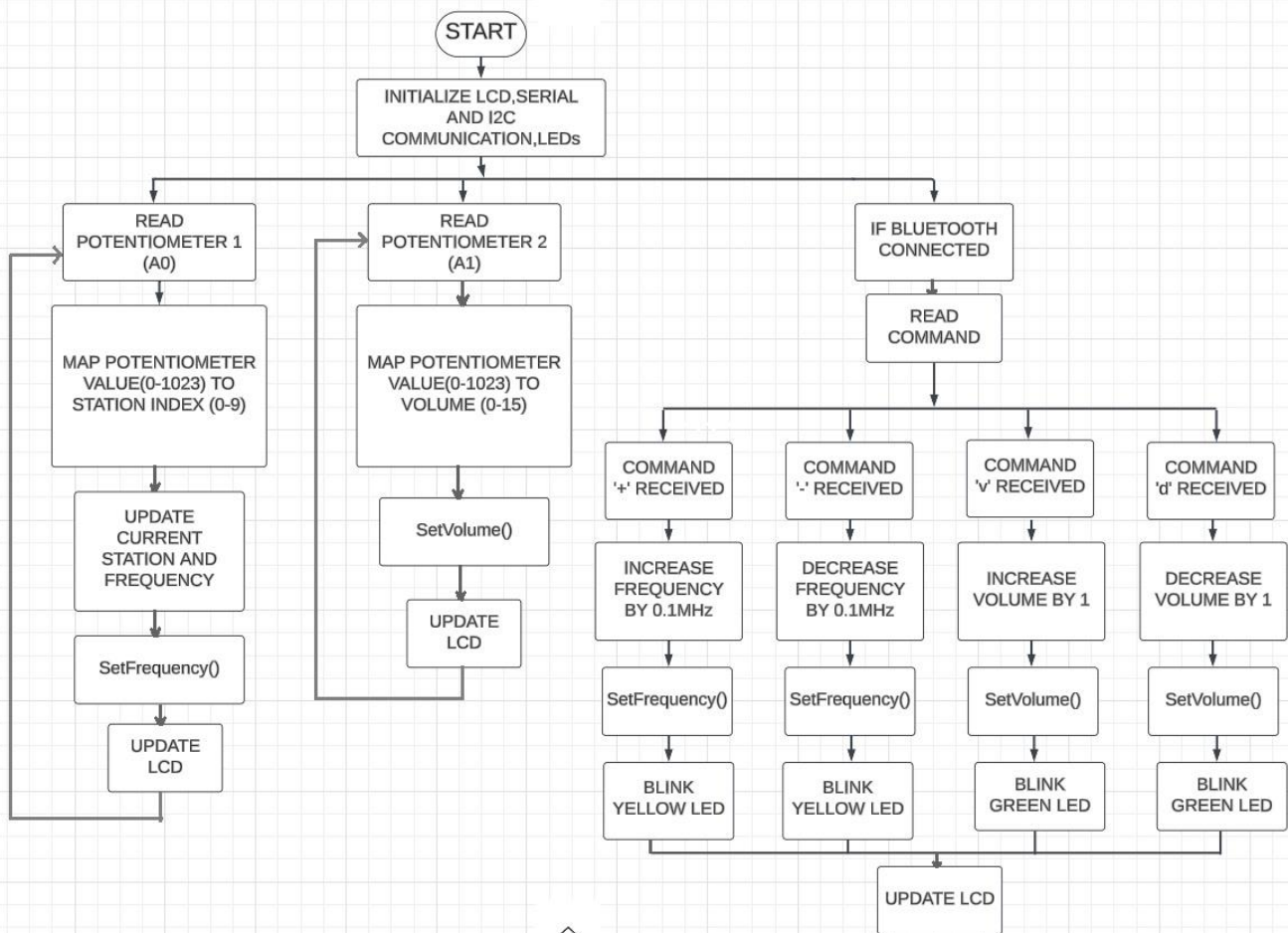


Figure 4.2.1 Flow chart

4.3 RADIO STATIONS

Table 4.1 Radio Stations

Station Index	Radio Station	City	Frequency (in MHz)
0	Radio City	Bangalore	91.1
1	Big FM	Bangalore	92.7
2	Red FM	Bangalore	93.5
3	Radio Mirchi	Bangalore	95
4	Radio Indigo	Bangalore	91.9
5	Fever FM	Bangalore	104
6	All India Rainbow	Bangalore	101.3
7	Radio One	Bangalore	94.3
8	AIR Vividh	Bangalore	102.9
9	All India Radio	Chitradurga	102.6

CHAPTER 5

HARDWARE DESCRIPTION

5.1 TEA5767 FM Radio Module



Figure 5.1.1 TEA5767 FM Radio Module

The **TEA5767 FM Radio Module** is a highly integrated, compact FM radio receiver that simplifies the process of building FM radio functionality into various projects. It operates in the FM frequency range and provides high-quality stereo audio output, making it ideal for portable and embedded systems.

Features

1. **Radio Receiver:** Covers the **FM frequency range of 76 MHz to 108 MHz**.
 - Provides **stereo audio output** for high-quality sound reproduction.
 - Integrated with phase-locked loop (PLL) circuitry for stable frequency reception.
2. **Interface:** Controlled via **I2C communication protocol**, making it easy to integrate with microcontrollers like Arduino, ESP32, or Raspberry Pi.
 - Compatible with standard I2C pins (SDA and SCL) for tuning and configuration.
3. **Low Power Consumption:** Designed for portable devices, ensuring minimal power usage to extend battery life.
4. **Onboard Features:** Includes built-in RF input, IF processing, and demodulation circuits.
 - Offers **soft mute** functionality to reduce background noise during weak signal reception.
 - Supports **automatic search tuning** for finding FM stations easily.



Figure 5.1.2 Interfacing diagram of TEA5767 IC

Interfacing of TEA5767 FM Radio module

1. Power Supply:

The TEA5767's VCC is connected to the Arduino Nano's 5V pin for power.

The GND pin is connected to the Arduino's GND to complete the circuit.

2. I²C Communication:

SDA (Serial Data Line): Connects to the Arduino's A4 pin for data transfer.

SCL (Serial Clock Line): Connects to the Arduino's A5 pin for clock synchronization.

3. Antenna:

An optional antenna (e.g., a 75 cm wire) can be connected to the ANT pin to improve FM signal reception.

5.2 Arduino Nano

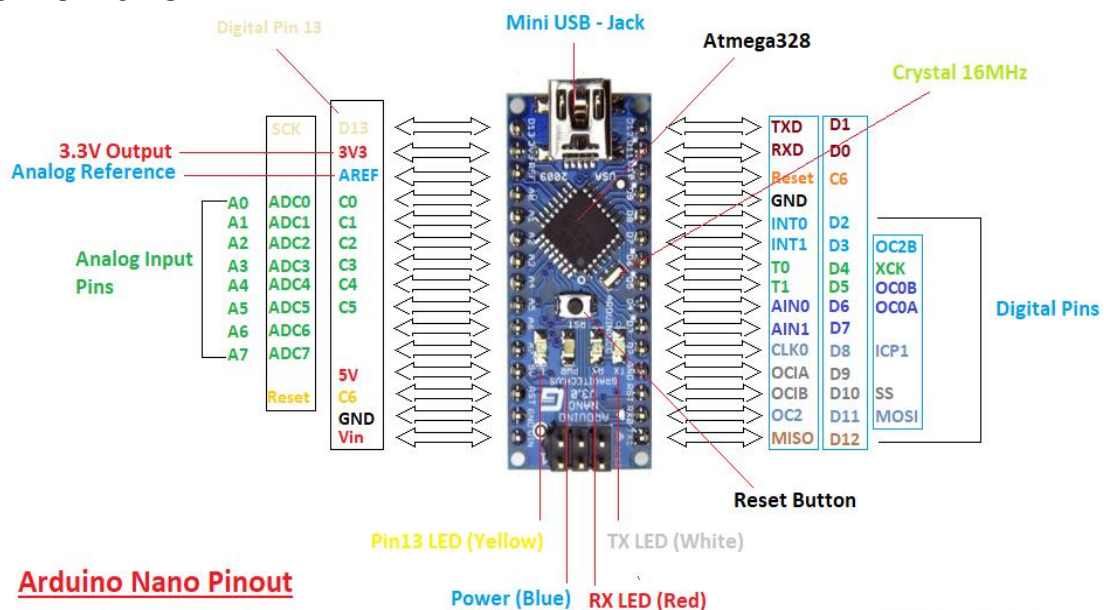


Figure 5.2.1 Interfacing diagram of Arduino Nano

The Arduino Nano is a compact, versatile microcontroller board based on the ATmega328P (or ATmega328P-AU in surface-mount). It is designed for embedded systems and prototyping, offering a small form factor with extensive I/O capabilities.

Features

1. Microcontroller Unit (MCU):

- **ATmega328P** (8-bit AVR microcontroller).

- Clock speed of **16 MHz**, providing sufficient processing power for most small to medium-scale applications.
- Operates at a voltage of **5V**.

2. Memory:

- **Flash Memory: 32 KB** (2 KB used by the bootloader).
- **SRAM: 2 KB**.
- **EEPROM: 1 KB**, useful for non-volatile storage.

3. I/O Capabilities:

- **14 Digital I/O Pins:** Pins D0-D13, 6 of which support PWM (Pulse Width Modulation).
- **8 Analog Inputs:** A0-A7 for reading analog signals.
- **PWM:** 6 pins (D3, D5, D6, D9, D10, D11) support PWM for applications like controlling LEDs and motors.

4. Communication Interfaces:

- **Serial Communication (UART):** Available on D0 (RX) and D1 (TX).
- **I2C:** Available on A4 (SDA) and A5 (SCL) for connecting to I2C devices like sensors.

5. Power Supply:

- Powered via **USB (5V)** or **Vin Pin (6-12V)** for external power.
- Onboard voltage regulators provide **5V** and **3.3V** outputs.
- Low power consumption with the ability to run on batteries for small projects.

6. Onboard LEDs:

- **Power LED:** Indicates the board is powered.
- **Pin 13 LED:** Used for debugging and testing.

5.3 Audio Amplifier LM386



Figure 5.3.1 Audio Amplifier LM386

The **LM386 Audio Amplifier** is a low-voltage power amplifier designed specifically for audio applications. Known for its compact size, low power consumption, and simplicity, the LM386 is an excellent choice for small and portable audio projects.

Features

1. Low Power Operation:

- Operates on a **wide voltage range from 4V to 12V**.
- Consumes minimal quiescent current, making it ideal for battery-operated devices.

2. Adjustable Gain:

- The amplifier has a default voltage gain of **20**, which can be increased up to **200** by adding an external capacitor and resistor.
- Provides flexibility in amplifying audio signals to meet the requirements of specific applications.

3. High Audio Quality:

- Offers clear audio amplification with minimal distortion.
- Capable of driving small speakers directly (up to **0.5W output**).

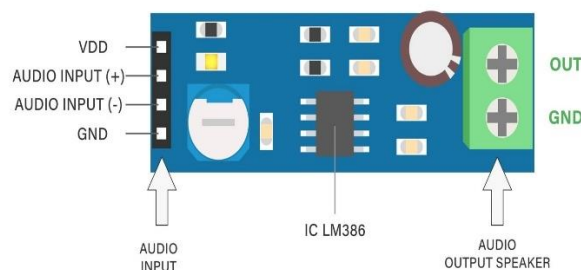


Figure 5.3.2 Interfacing diagram of LM386 Audio Amplifier

Interfacing of LM386 Audio Amplifier

1. Power Supply:

VCC: Connect to the 5V pin on the Arduino Nano (or 9V if you want higher power output).

GND: Connect to the GND of both the Arduino Nano and the TEA5767 module to complete the circuit.

2. Audio Input:

The **IN** pin of the LM386 connects to the audio output pin of the **TEA5767**. This carries the FM radio signal from the TEA5767 that will be amplified by the LM386.

3. Audio Output:

OUT: The LM386's audio output (OUT pin) is connected to the speaker. This will drive the sound output.

5.4 Potentiometer



Figure 5.4.1 10k Potentiometer

A **potentiometer** is a three-terminal variable resistor widely used to control electrical parameters such as voltage and resistance. It is a simple yet versatile component that can adjust parameters like volume, brightness, or frequency in electronic circuits.

Features

1. Adjustable Resistance:

- Potentiometers allow resistance to be adjusted manually by rotating a knob or sliding a lever.
- Common resistance values include **10k Ω** , **50k Ω** , and **100k Ω** , making them suitable for a wide range of applications.

2. Types:

- **Rotary Potentiometers:** Circular knob that rotates to adjust resistance.
- **Linear Potentiometers:** Slider mechanism that moves linearly to vary resistance.
- **Trimmer Potentiometers:** Small, adjustable potentiometers typically used for fine-tuning in circuits.

5.5 HC-05 Bluetooth Module

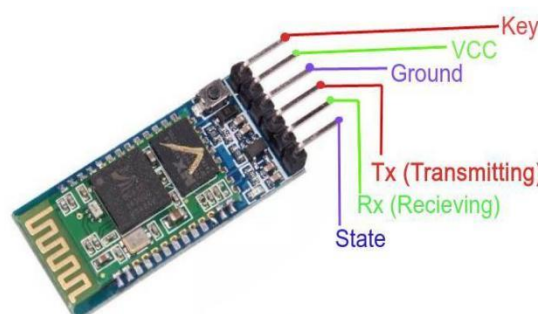


Figure 5.5.1 Interfacing diagram of HC-05 Bluetooth Module

The **HC-05** is a widely used Bluetooth module designed for wireless communication in embedded systems and IoT projects. It operates using Bluetooth 2.0 technology, enabling seamless serial communication between microcontrollers and Bluetooth-enabled devices like smartphones, laptops, or other Bluetooth modules.

Features

1. **Bluetooth Protocol:** Operates on **Bluetooth 2.0 with Enhanced Data Rate (EDR)**.
 - Supports both **Master** and **Slave modes**, allowing flexibility in communication roles.
2. **Communication Interface:** Communicates via a standard **UART (Universal Asynchronous Receiver-Transmitter)** interface.
 - Configurable baud rates (default is 9600 bps) for serial communication.
3. **Operating Voltage:** Works within a voltage range of 3.3V to 6V, making it compatible with most microcontrollers.
 - Includes an onboard voltage regulator for safe operation.
4. **Range and Speed:** Offers a wireless range of up to 10 meters in open space.
 - Data transfer speed up to **3 Mbps**, suitable for most applications.
5. **Configurable Parameters:** Features an AT command mode for customizing device settings, such as device name, baud rate, and role (Master/Slave).

Interfacing of HC-05 Bluetooth module

1. **Power Supply:** Connect the VCC of the HC-05 to the 5V of the Arduino Nano.
Connect the GND of the HC-05 to the GND of the Arduino Nano.
2. **Communication:** TXD (Transmit) of the HC-05 connects to the RX (pin D0) of the Arduino Nano (through a voltage divider to step down voltage).
RXD (Receive) of the HC-05 connects to the TX (pin D1) of the Arduino Nano.

5.6 16x2 I2C LCD Display



Figure 5.6.1 Interfacing diagram of 16x2 I2C LCD Display

The 16x2 I2C LCD display is a commonly used alphanumeric display module in embedded systems. It features a 16-character, 2-line format and utilizes the I2C interface, simplifying connections and reducing the number of pins required for communication with microcontrollers.

Features

1. **Display Type:** Alphanumeric display capable of showing 16 characters per line over 2 lines.
 - Supports a 5x8 dot matrix for each character, with the ability to create custom characters.
2. **Interface:** Operates via the I2C (Inter-Integrated Circuit) protocol for efficient communication.

- Requires only two pins (SDA and SCL) for data transmission, in addition to power connections.
- 3. Operating Voltage: Operates at 5V DC (or 3.3V DC for some models), making it compatible with a wide range of microcontrollers.
- Includes an onboard I2C adapter for seamless integration.
- 4. Backlight and Contrast: Equipped with an LED backlight for enhanced visibility in low-light conditions.
- Contrast can be adjusted via a potentiometer or software control.
- 5. Compact Design: Compact dimensions (~80mm x 35mm x 12mm), ideal for embedded projects with limited space.
- Lightweight and easy to mount.
- 6. Ease of Use: Preloaded with HD44780 or compatible controller IC for simplified operation.
- Supported by popular microcontroller libraries (e.g., Arduino LiquidCrystal_I2C library) for easy programming.
- 7. Performance: Capable of updating text efficiently, with a response suitable for real-time applications.
- Default I2C address: 0x27 or 0x3F (configurable).

Interfacing of 16x2 LCD display

1. Power Supply:

VCC: Connect to the 5V pin of the Arduino Nano to power the LCD.

GND: Connect to the GND pin of the Arduino Nano to complete the circuit.

2. I²C Communication:

SDA (Serial Data): Connect to the A4 pin on the Arduino Nano (used for I²C data transfer).

SCL (Serial Clock): Connect to the A5 pin on the Arduino Nano (used for I²C clock signal).

5.7 Circuit Diagram

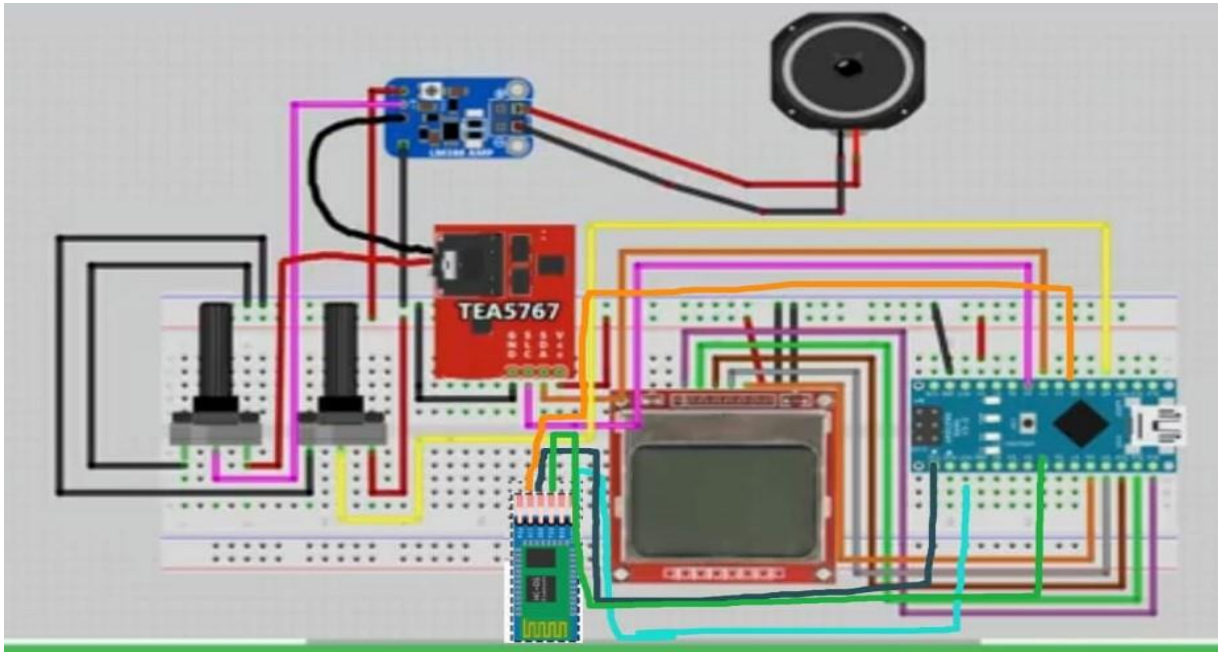


Figure 5.7.1 Schematic Diagram of Smart FM Radio Receiver using Arduino Nano

The circuit is built around an **Arduino Nano**, which serves as the primary controller, managing the communication and data flow between all the components. The system includes a **TEA5767 FM Radio Module**, which is connected to the Arduino using the **I²C protocol** (SDA to A4, SCL to A5). This module handles FM signal reception, while its output is sent to an **LM386 audio amplifier**. The amplified audio signal drives the connected **speaker**, delivering clear sound output.

Two **potentiometers** are used in the circuit—one to control the **volume** and the other to adjust the **radio frequency**. These potentiometers send analog signals to the Arduino, allowing smooth and user-friendly control over audio settings. The **HC-05 Bluetooth module** is also integrated for wireless communication, enabling remote control of the system via a mobile application.

CHAPTER 6

SOFTWARE DESCRIPTION

The Smart Radio Project leverages various software tools and components to develop, debug, and enhance the functionality of the system. These tools ensure efficient communication between hardware modules, a seamless user interface, and ease of customization for additional features.

6.1 Wokwi Simulator (Simulation)

Wokwi is an advanced online simulation platform for prototyping and testing embedded projects without physical hardware. It allows users to simulate microcontroller-based circuits, write code, and visualize the functionality of various components in a virtual environment.

Features

1. **Circuit Simulation:** Provides a drag-and-drop interface for designing circuits with microcontrollers, sensors, displays, and other components.
 - Real-time visualization of the circuit's operation.
2. **Code Editor:** Includes an integrated code editor with support for Arduino programming.
 - Features syntax highlighting, code completion, and error detection.
3. **Hardware Component Library:** Offers a wide range of components, such as microcontrollers, LCD displays, potentiometers, and Bluetooth modules, for circuit simulation.
 - Components behave like their real-world counterparts, ensuring accurate simulation results.
4. **Serial Monitor:** Includes a built-in serial monitor for debugging and monitoring serial communication during the simulation.
5. **Cross-Platform Access:** Accessible through any modern web browser, with no installation required.
 - Supports seamless collaboration and sharing of projects via sharable links.

Usage in Project

1. **Simulating the Circuit**
 - The circuit, including the microcontroller, LCD, push buttons, potentiometers, and LEDs are assembled and tested virtually on Wokwi.
2. **Code Testing**
 - The project code is uploaded to the simulated microcontroller, enabling verification of functionalities like frequency tuning and volume control.
3. **Debugging and Refinement**
 - The serial monitor and real-time simulation help in identifying and rectifying bugs in the code.
 - Allows testing of different configurations without hardware constraints.
4. **Documentation and Sharing**
 - Generates visual representations of the circuit and code for documentation purposes.
 - Projects can be shared via links for collaboration or demonstration.

6.2 Arduino IDE

The **Arduino Integrated Development Environment (IDE)** is the main platform used to write, compile, and upload the code to the Arduino Nano. It offers a user-friendly interface and simplifies microcontroller programming.

Features:

- **Code Editor:** Provides syntax highlighting, automatic formatting, and error checking.
- **Serial Monitor:** Used for debugging and monitoring runtime messages sent by the Arduino Nano, such as the current frequency or volume changes.
- **Library Management:** Includes a library manager to easily import and manage additional libraries, such as `Wire.h` for I2C communication and `LiquidCrystal_I2C.h` for the LCD display.
- **Upload Tool:** Allows direct uploading of the compiled program to the Arduino Nano via a USB connection.

Usage in Project:

- Writing and uploading the code to the Arduino Nano.
- Debugging the program by using the Serial Monitor to check outputs like frequency and volume changes or Bluetooth commands received.

Libraries and Modules

The project uses multiple Arduino libraries that simplify the interaction with hardware components and ensure reliable functionality.

1. **Wire.h:**

- Facilitates I2C communication between the Arduino Nano and the TEA5767 FM Radio Module.
- Used to send frequency and volume data to the TEA5767.

2. **LiquidCrystal_I2C.h:**

- Enables interaction with the I2C-based LCD display.
- Used to display the current station name, frequency, and volume level dynamically.

3. **SoftwareSerial.h** (if used for HC-05 Bluetooth module):

- Allows the Arduino Nano to communicate with the HC-05 Bluetooth module via software-implemented serial communication.
- Useful when the Arduino Nano's default hardware serial port is reserved for debugging.

6.3 Bluetooth Electronics App

The **Bluetooth Electronics** app is used for controlling the radio remotely via the HC-05 Bluetooth module.

Features:

- Customizable buttons and sliders to send specific commands (+, -, v, d) to control the frequency and volume.
- Real-time feedback on commands sent, ensuring the user can monitor and adjust settings wirelessly.

Usage in Project:

- Commands from the app are received by the HC-05 Bluetooth module and processed by the Arduino Nano to:
 - Adjust frequency (+ or -).
 - Increase or decrease volume (v or d).
- Provides a user-friendly interface for wireless control.

CHAPTER 7

RESULTS AND DISCUSSIONS

7.1 Simulation:

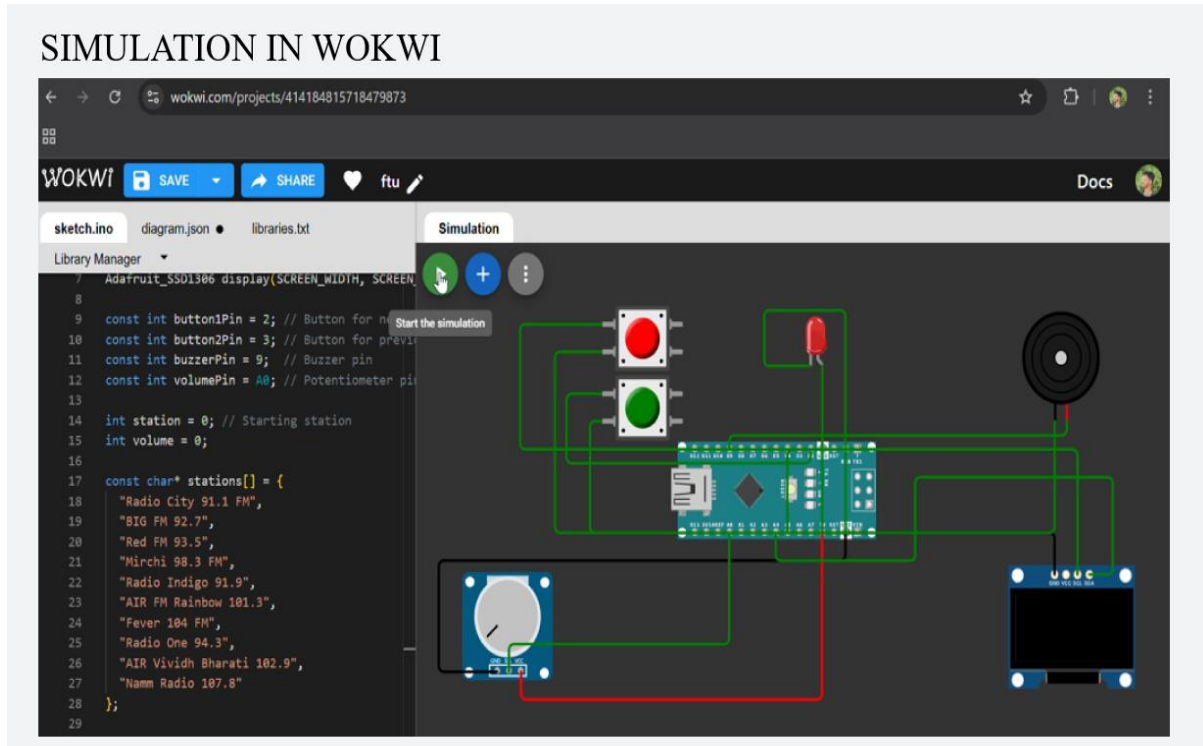


Figure 7.1.1 Simulation in wokwi

The simulation demonstrates a **basic smart radio system** with the following functionalities:

1. Station Selection:

- Two push buttons are used to navigate through a list of predefined radio stations.
- One button increments the station to the next one, while the other decrements it to the previous one.
- The station list loops back to the start after the last station and vice versa.

2. Volume Control:

- A potentiometer is used to simulate volume adjustment.
- Rotating the potentiometer changes the volume, which is converted into a decibel (dB) level.

3. Audio Output Simulation:

- A buzzer is used to mimic audio output.
- The tone of the buzzer changes depending on the volume, creating a simple audio feedback system.

4. Real-Time Interaction:

- Users can interact with buttons and the potentiometer to switch stations and adjust the volume in real time.
- The simulation provides an intuitive way to understand how the components work together in a physical setup.

This simulation serves as a practical demonstration of integrating input devices (buttons, potentiometer) and an output device (buzzer) with an Arduino Nano for a functional smart radio prototype.

Simulation Result:

Station	Volume	Display Output	Buzzer sound (in dB)
Radio city 91.1 FM	0	Station:Radio city Volume:0	0 dB (Silent)
Red FM 93.5	50	Station: Red FM 93.5 Volume: 50	~50 dB 40 to 60dB (medium pitch sound)
Radio Mirchi 98.3 FM	20	Station: Radio Mirchi 98.3 FM Volume: 20	~30 dB 10 to 40dB (low pitch sound)
Fever 104 FM	75	Station: Fever 104 FM Volume: 75	~65 dB 60 to 70dB (high pitch sound)
All India Radio 102.6 FM	102	Station: All India Radio 102.6 FM Volume: 102	~80 dB 70 to 80dB (highest pitch sound)

Table 7.1.1 Simulation Tabulation

7.2 RESULTS OF PROJECT:

1]



Figure 7.2.1 Volume control via Bluetooth

The volume of the output signal was changed as per the user input. Users could adjust the volume using Bluetooth commands ('v' to increase and 'd' to decrease), with the current volume level displayed temporarily on the LCD for clarity. The system allowed volume levels to range from 0 (mute) to 15 (maximum), ensuring a comfortable audio experience. The green LED provided additional feedback by blinking with each volume adjustment, making the interaction intuitive and user-friendly.

2]



Figure 7.2.2 Tuning to different Radio Station

The station selection feature worked efficiently through the potentiometer and Bluetooth commands. The potentiometer allowed users to switch between predefined stations, with smooth transitions and no signal interruptions. For fine-tuning, Bluetooth commands ('+' to increase frequency and '-' to decrease) enabled precise control, allowing users to lock onto stations with weak signals. The yellow LED blinked to confirm frequency changes, and the LCD updated in real-time, displaying the station name and frequency, ensuring an engaging and interactive experience.

CHAPTER 8

APPLICATIONS

1. Personal Entertainment System

- Offers a compact and customizable device for enjoying music and radio channels, tailored to specific tastes and preferences.
- Supports integration with Spotify and TuneIn for modern digital streaming.

2. Educational Tool

- Demonstrates principles of radio communication, frequency modulation, and Arduino programming.
- Useful for teaching concepts like wireless communication and audio amplification in academic settings.

3. Home Automation

- Can be part of a smart home system to provide audio notifications or background music.
- Bluetooth control allows for seamless integration with mobile devices.

4. Portable Music Player

- Its compact size and low power consumption make it an excellent choice for portable audio solutions.
- Customizable radio stations ensure access to favourite local channels on the go.

5. Assistive Technology

- Simplified control buttons make it user-friendly for elderly or differently-abled individuals.
- Bluetooth connectivity enables remote operation for those with mobility challenges.

6. DIY Learning and Hobbies

- Encourages hands-on learning for hobbyists interested in electronics and IoT.
- Provides a starting point for integrating other IoT features like voice assistants or smart displays.

7. Community Use

- Can serve as a community radio device, broadcasting local or curated content in small settings.
- Useful in remote areas where digital radio integration enhances the variety of content.

CHAPTER 9

ADVANTAGES

1. Cost-Effective

- Built using affordable components like the TEA5767 radio module, Arduino Nano, and LM386 amplifier, making it budget-friendly compared to commercial alternatives.

2. Compact and Portable

- The small form factor makes it ideal for personal use and easy to carry around.
- Low power consumption ensures extended usage with minimal energy requirements.

3. Versatile Connectivity

- Bluetooth integration allows remote control via mobile devices, enhancing convenience.
- Potential to integrate with other IoT systems for added functionality.

4. Improved User Experience

- User-friendly interface with potentiometers and virtual controls simplifies operation.

5. Educational Value

- Serves as a learning platform for understanding electronic circuits, programming, and audio processing.
- Demonstrates integration of hardware (radio module, potentiometers, amplifier) and software (Arduino coding).

6. Localized Content Access

- Enables access to specific Bangalore radio stations or any location-based content, making it highly relevant for regional use.

7. Eco-Friendly

- Uses energy-efficient components and has potential for solar-powered or battery operation, reducing its environmental impact.

LIMITATIONS

1. Limited Audio Quality

- The use of an LM386 amplifier and a 0.5-watt speaker may not provide high-fidelity audio output, especially at higher volumes.

2. Environmental Interference

- FM radio reception can be affected by environmental factors, leading to signal distortion or loss in certain areas.

3. Regional Limitation

- FM radio stations are specific to regions like Bangalore, so users relocating may have to reconfigure station presets.

CONCLUSION

The smart FM radio project is an innovative and versatile solution that bridges the gap between traditional FM radio systems and modern digital technologies. By integrating features like user-friendly controls, the project addresses the growing demand for customizable and practical devices. It provides an efficient and compact system suitable for both personal and educational use.

While the project has some limitations, such as basic audio quality but it excels in portability, ease of operation, and flexibility. It also presents significant potential for future upgrades, including enhanced audio performance, additional digital features, and further integration with IoT systems. Overall, this project showcases the capability of combining traditional and modern technologies to create a functional, affordable, and scalable device, serving as a valuable platform for learning and practical applications.

FUTURE SCOPE

Future Scope of the Smart FM Radio Receiver are:

1. Enhanced Connectivity

- Integration with Wi-Fi: Adding Wi-Fi connectivity for streaming internet radio stations, podcasts, or other online audio content.
- Voice Control: Incorporating voice assistants like Google Assistant or Alexa for hands-free operation.

2. Expanded User Interface

- Touchscreen Integration: Upgrading the interface to a touchscreen for intuitive navigation and controls.
- Mobile App Development: Creating a dedicated mobile app for more advanced controls, station selection, and volume adjustments.

3. Improved Audio Experience

- Stereo Output: Upgrading the audio system to support stereo sound for better audio quality.
- DSP Integration: Adding Digital Signal Processing (DSP) for enhanced audio clarity, equalization, and noise reduction.

4. Power Optimization

- Battery Support: Incorporating a rechargeable battery with power management for portability.
- Low-Power Operation: Optimizing software and hardware to reduce power consumption.

5. Additional Functionalities

- Station Presets with Memory: Allowing users to save and recall favorite stations even after a power reset.
- RDS (Radio Data System): Enabling the display of station information, song titles, and artist names via RDS.
- Alarm Clock Feature: Adding functionality to set alarms with radio playback.

6. IoT Integration

- Smart Home Compatibility: Enabling the device to connect with smart home systems for centralized control.
- Cloud Connectivity: Storing and syncing user preferences through cloud services.

7. Multi-Language Support

- Dynamic Display Options: Supporting multiple languages on the LCD for station names, commands, and menus to cater to a global audience.

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APPENDIX-A: Data Sheets

DATA SHEET OF ARDUINO NANO

Section	Description
Microcontroller	ATmega328P
Operating Voltage	5V
Input Voltage	7-12V
Digital I/O pins	14
PWM I/O pins	6
Analog Input pins	8
DC current per I/O pins	40 mA
Flash memory	32KB
SRAM	2KB
EEPROM	1KB
Clock Speed	16 MHz
LED_BUILTIN	13
Dimensions	45mmx18mm
Weight	7g
VIN	External power input(7-12V)
5V	Regulated 5V output
3.3V	Regulated 3.3V output
GND	Ground pins
Serial Communication	Pins 0 (RX) and 1 (TX)
I2C Communication	A4 (SDA) and A5 (SCL)
SPI Communication	Pins 10,11,12,13

DATA SHEET OF TEA5767 FM RADIO MODULE

Section	Description
Frequency Range	76 MHz to 108 MHz
Supply Voltage	2.7V to 5.5V
Current Consumption	10 mA
Interface	I2C (SDA and SCL)
Sensitivity	High sensitivity with low-noise RF input
Output	Stereo audio output
Dimensions	25mmx20mm

DATA SHEET OF LM386 AUDIO AMPLIFIER

Section	Description
Operating Voltage	4V to 15V
Power Output	0.5W
Input Impedance	50 k-Ohm
Voltage Gain	20 to 200
Output Load Impedance	4 Ohm to 32 Ohm
Current Consumption	4mA
Distortion	Less than 1%

APPENDIX-B: CODE

Simulation code

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

const int button1Pin = 2; // Button for next station
const int button2Pin = 3; // Button for previous station
const int buzzerPin = 9; // Buzzer pin
const int volumePin = A0; // Potentiometer pin for volume control

int station = 0; // Starting station
float volumeDB = 0.0; // Volume in decibels

const char* stations[] = {
  "Radio City 91.1 FM",
  "BIG FM 92.7",
  "Red FM 93.5",
  "Mirchi 98.3 FM",
  "Radio Indigo 91.9",
  "AIR FM Rainbow 101.3",
  "Fever 104 FM",
  "Radio One 94.3",
  "AIR Vividh Bharati 102.9",
  "Namm Radio 107.8"
};

void setup() {
  pinMode(button1Pin, INPUT_PULLUP);
  pinMode(button2Pin, INPUT_PULLUP);
  pinMode(buzzerPin, OUTPUT);
  Serial.begin(9600);
```

```

// Initialize OLED display
if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("SSD1306 allocation failed"));
    for (;;)
    }
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(WHITE);
display.display();

// Show initial station
showStation();
}

void loop() {
    // Check buttons for station change
    if (digitalRead(button1Pin) == LOW) {
        station++;
        if (station > 9) station = 0; // Loop back to the first station
        showStation();
        delay(300); // Debounce delay
    }
    if (digitalRead(button2Pin) == LOW) {
        station--;
        if (station < 0) station = 9; // Loop back to the last station
        showStation();
        delay(300); // Debounce delay
    }

    // Read potentiometer for volume control
    int volumeRaw = analogRead(volumePin);
    volumeDB = calculateDecibels(volumeRaw);

    showStation(); // Update display with volume in dB
    playAudio(volumeDB);
}

```

```

void showStation() {
    display.clearDisplay();
    display.setCursor(0, 0);
    display.print("Smart Radio");
    display.setCursor(0, 20);
    display.print("Station: ");
    display.println(stations[station]);
    display.setCursor(0, 40);
    display.print("Volume: ");
    display.print(volumeDB, 1); // Display volume with 1 decimal place
    display.print(" dB");
    display.display();
}

void playAudio(float volumeLevel) {
    if (volumeLevel > 0) {
        tone(buzzerPin, 1000 + (volumeLevel * 10), 200); // Adjust frequency based on volume
        delay(200); // Delay to simulate audio duration
    } else {
        noTone(buzzerPin); // Mute if volume is zero
    }
}

float calculateDecibels(int rawValue) {
    // Map raw value (0-1023) to decibels (0 to -50 dB as an example)
    if (rawValue == 0) return -50.0; // Avoid log(0) error
    float maxRaw = 1023.0;
    return 20.0 * log10(rawValue / maxRaw); // Formula: 20 * log10(value / max)
}

```

Source code

```

#include <Wire. h>
#include <LiquidCrystal_I2C.h>

// TEA5767 I2C address
#define TEA5767_ADDRESS 0x60

// Initialize the I2C LCD (Address: 0x27, 16 columns, 2 rows)
LiquidCrystal_I2C lcd(0x27, 16, 2);

// LED pins
const int yellowLED = 2; // Frequency change LED
const int greenLED = 3; // Volume change LED

// Define an array of 10 radio stations with name and frequency
struct Station {
    String name;
    float frequency;
};

Station stations[10] = {
    {"Radio City", 91.1},
    {"Radio Indigo", 91.9},
    {"Big FM", 92.7},
    {"Red FM", 93.5},
    {"Radio One", 94.3},
    {"Radio Mirchi", 95.0},
    {"AllIndiaRainbow", 101.3},
    {"All India Radio", 102.6},
    {"AIRVividhBharti", 102.9},
    {"Fever FM", 104.0}
};

int currentStation = 4; // Start with the station at index 4 (Frequency: 96.0)
float frequency = 96.0; // Default frequency
int volume = 8;         // Volume level (range: 0 to 15)

```

```

void setup() {
  // Initialize Serial Monitor
  Serial.begin(9600);

  // Initialize the LCD
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("FM Radio");
  lcd.setCursor(0, 1);
  lcd.print("Initializing...");
  delay(2000);

  // Initialize I2C
  Wire.begin();

  // Initialize LEDs
  pinMode(yellowLED, OUTPUT);
  pinMode(greenLED, OUTPUT);

  // Set initial station
  setFrequency(stations[currentStation].frequency);

  // Update display with the current station
  updateDisplay();
}

void loop() {
  // Potentiometer for preset stations
  int potValue = analogRead(A0); // Read the potentiometer value (0-1023)
  int newStation = map(potValue, 0, 1023, 0, 9);

  if (newStation != currentStation) {
    currentStation = newStation;
    frequency = stations[currentStation].frequency;
    setFrequency(frequency);
  }
}

```

```

    updateDisplay();
    blinkLED(yellowLED);
}

// Bluetooth commands for fine-tuning and volume
if (Serial.available()) {
    char command = Serial.read();

    if (command == '+') {
        frequency += 0.1; // Increase frequency
        setFrequency(frequency); // Do not call syncStationWithFrequency here===pppppppppppp-
        blinkLED(yellowLED);
    } else if (command == '-') {
        frequency -= 0.1; // Decrease frequency
        setFrequency(frequency); // Do not call syncStationWithFrequency here
        blinkLED(yellowLED);
    } else if (command == 'v') {
        volume = min(volume + 1, 15); // Increase volume
        setVolume(volume);
        showTemporaryVolume();
        blinkLED(greenLED);
    } else if (command == 'd') {
        volume = max(volume - 1, 0); // Decrease volume
        setVolume(volume);
        showTemporaryVolume();
        blinkLED(greenLED);
    }
    updateDisplay(); // Update display with new values
}

delay(200); // Add delay for stability
}

void setFrequency(float freq) {
    uint8_t frequencyB[5];
    uint16_t freqB = (freq * 1000000 + 225000) / 32768;

    frequencyB[0] = freqB >> 8;

```

```

frequencyB[1] = freqB & 0xFF;
frequencyB[2] = 0xB0; // High side injection
frequencyB[3] = 0x10; // Stereo
frequencyB[4] = 0x00;

Wire.beginTransaction(TEA5767_ADDRESS);
for (int i = 0; i < 5; i++) {
    Wire.write(frequencyB[i]);
}
Wire.endTransmission();
Serial.print("Frequency set to: ");
Serial.println(freq, 1);
}

void setVolume(int vol) {
    Wire.beginTransaction(TEA5767_ADDRESS);
    Wire.write(0xFF); // Fake bytes to simulate volume control
    Wire.write(vol);
    Wire.endTransmission();
    Serial.print("Volume set to: ");
    Serial.println(vol);
}

void updateDisplay() {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print(stations[currentStation].name); // Display station name
    lcd.setCursor(0, 1);

    lcd.print("Freq: ");
    lcd.print(frequency, 1); // Display frequency with 1 decimal
    lcd.print(" MHz");
}

void showTemporaryVolume() {
    // Display the volume level for a short duration
    lcd.clear();

```



```

lcd.setCursor(0, 0);
lcd.print("Volume: ");
lcd.print(volume);
delay(1000); // Show volume for 1 second
updateDisplay(); // Revert to the main display
}

```

```

void blinkLED(int ledPin) {
    digitalWrite(ledPin, HIGH); // Turn on the LED
    delay(200);                // Wait for 200 ms
    digitalWrite(ledPin, LOW); // Turn off the LED
}

```

```

void syncStationWithFrequency() {
    // This function is not called when fine-tuning the frequency anymore.
    // Now it only matches the station when explicitly changing the station via potentiometer or
    Bluetooth.
}

```

Icons.c

```

#if defined(__AVR__)
    #include <avr/pgmspace.h>
    #define imagedatatype const uint8_t
#elif defined(__PIC32MX__)
    #define PROGMEM
    #define imagedatatype const unsigned char
#elif defined(__arm__)
    #define PROGMEM
    #define imagedatatype const unsigned char
#endif

```

```

imagedatatype signal5[] PROGMEM={
0xC1, 0xC3, 0xC5, 0xF9, 0xC5, 0xC3, 0xC1, 0xC0, 0xE0, 0xC0, 0xF0, 0xC0, 0xF8, 0xC0, 0xFC,
0xC0, // 0x0010 (16) pixels
0xFE,
};
imagedatatype signal4[] PROGMEM={

```

```
0xC1, 0xC3, 0xC5, 0xF9, 0xC5, 0xC3, 0xC1, 0xC0, 0xE0, 0xC0, 0xF0, 0xC0, 0xF8, 0xC0, 0xFC,
0xC0, // 0x0010 (16) pixels
0xC0,
};
```

```
imagedatatype signal3[] PROGMEM={
0xC1, 0xC3, 0xC5, 0xF9, 0xC5, 0xC3, 0xC1, 0xC0, 0xE0, 0xC0, 0xF0, 0xC0, 0xF8, 0xC0, 0xC0,
0xC0, // 0x0010 (16) pixels
0xC0,
};
```

```
imagedatatype signal2[] PROGMEM={
0xC1, 0xC3, 0xC5, 0xF9, 0xC5, 0xC3, 0xC1, 0xC0, 0xE0, 0xC0, 0xF0, 0xC0, 0xC0, 0xC0, 0xC0,
0xC0, // 0x0010 (16) pixels
0xC0,
};
```

```
imagedatatype signal1[] PROGMEM={
0xC1, 0xC3, 0xC5, 0xF9, 0xC5, 0xC3, 0xC1, 0xC0, 0xE0, 0xC0, 0xC0, 0xC0, 0xC0, 0xC0, 0xC0,
0xC0, // 0x0010 (16) pixels
0xC0,
};
```

Splash.c

```
#include <avr/pgmspace.h>
```

```
const unsigned char splash [] PROGMEM = {
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x80, 0x80, 0x80, 0x80, 0xC0, 0xC0,
0x40, 0x60, 0x60, 0x20, 0x20, 0x30, 0x10, 0x10, 0x1C, 0x3E, 0x3E, 0x3E, 0x1C, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80,
0x00, 0x80, 0x80, 0x80, 0x80, 0x00, 0x00, 0x00, 0x00, 0x80, 0x80, 0x80, 0x80, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x80, 0xE0, 0xF0, 0xF0, 0xF0,
0xF0, 0xF0, 0x98, 0x98, 0x98, 0x8C, 0x8C, 0x8C, 0x8C, 0x86, 0x86, 0x86, 0x83, 0x83, 0x83,
0x81, 0x81, 0x81, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80, 0x80,
```

```

0x80, 0x80, 0x80, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0xFF, 0xFF,
0xFF, 0xE3, 0xE3, 0xE3, 0xE3, 0x03, 0x03, 0xE0, 0xFF, 0xFF, 0x7F, 0x7F, 0xFC, 0xE0, 0xC0,
0xF8, 0xFF, 0x1F, 0xFF, 0xFF, 0xFC, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0xFC, 0xFF, 0xFF, 0xFF, 0xFF, 0x1F, 0x0F, 0x0F, 0x0F, 0x1F, 0xBF, 0xFF, 0xFF, 0xFF, 0xFF,
0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x7F, 0x3F, 0x3F, 0x1F, 0x1F, 0x0F, 0x4F, 0x4F,
0x4F, 0x4F, 0x0F, 0x1F, 0x1F, 0x3F, 0x7F, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFC,
0x00, 0x00, 0x80, 0x80, 0x80, 0x80, 0x80, 0x0F, 0x0F, 0x0F, 0x00, 0x00, 0x80, 0x80, 0x00,
0x00, 0x0F, 0x0F, 0x8F, 0x80, 0x80, 0x87, 0x0F, 0x0F, 0x0F, 0x00, 0x80, 0x8F, 0x0F, 0x0F,
0x00, 0x80, 0x80, 0x80, 0x80, 0x00, 0x00, 0x00, 0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x3F,
0x1E, 0x1E, 0x1E, 0x1E, 0x7F, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x9F,
0x01, 0x00, 0x24, 0x26, 0x26, 0x26, 0x26, 0x26, 0x26, 0x26, 0x26, 0x26, 0x26, 0x26, 0x24,
0x20, 0x00, 0x03, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0x00, 0x00, 0x3F, 0x3F, 0x0D, 0x1D, 0x3F,
0x27, 0x00, 0x20, 0x38, 0x1F, 0x1B, 0x1F, 0x3E, 0x38, 0x00, 0x00, 0x3F, 0x3F, 0x31, 0x31,
0x1F, 0x0E, 0x00, 0x00, 0x3F, 0x3F, 0x00, 0x0E, 0x1F, 0x31, 0x31, 0x31, 0x31, 0x1F, 0x0E,
0x00, 0x00, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFE, 0xFC, 0xFC, 0xFC, 0xFE, 0xFF, 0xFF, 0xFF,
0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFC, 0xF0, 0xE1, 0xE3, 0xC3, 0x83, 0x93,
0x93, 0x93, 0x93, 0x93, 0x93, 0x83, 0xC3, 0xC3, 0xE3, 0xF0, 0xF8, 0xFE, 0xFF, 0xFF, 0xFF,
0xFF, 0xFF, 0x00, 0x00, 0x00, 0x00, 0x00, 0x40, 0xC0, 0x00, 0x00, 0xC0, 0xC0, 0x00, 0x00,
0x00, 0x60, 0xE0, 0xE0, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x80, 0xC0, 0x60, 0x60,
0x60, 0xC0, 0x80, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x01, 0x07, 0x0F, 0x0F,
0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F,
0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F,
0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x0F, 0x07, 0x01, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x03, 0x0F, 0x0E, 0x07, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x0F, 0x0F, 0x00, 0x00,
0x0C, 0x0C, 0x00, 0x00, 0x00, 0x03, 0x07, 0x0C, 0x0C, 0x0C, 0x07, 0x03, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00,
};

```

COURSE OUTCOMES

On completion of this course, students are able to

- 1) Solve the identified problems.
- 2) Analyze the available resources and their utilization.
- 3) Present the work carried out and prepare the report.

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